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# Bending Responses of The Human Cervical Spine

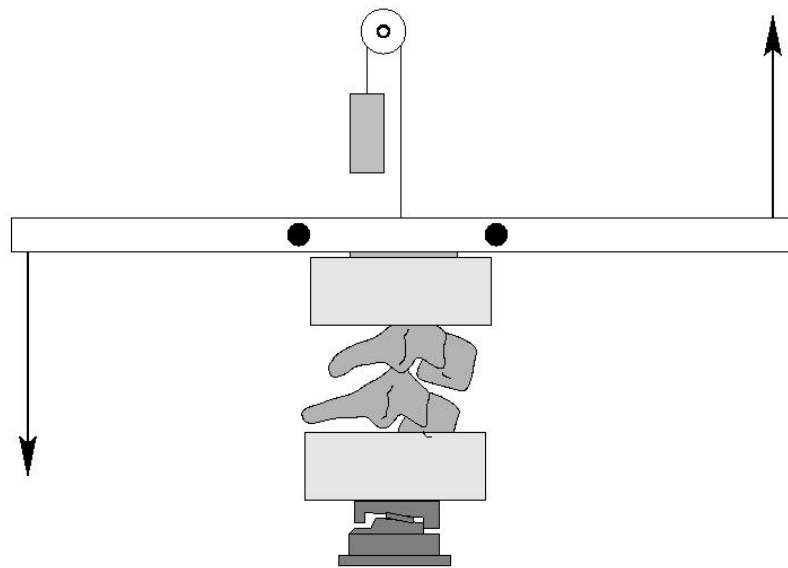
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## Flexibility Test Report

Test Reference: B35MO2EXTN

Test Date: 2/27/2003

Submission Date: 11/1/2003



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Duke University, Department of Biomedical Engineering  
NHTSA Cooperative Agreement No. *DTNH22-94-Y-07133*

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## Study Objective:

The objective of this study is to provide data on the responses of the human cervical spine to pure bending. The experimental protocol consists of three tests: two flexibility tests and a failure test. The flexibility tests are designed to quantify the static response of motion segments to flexion and extension moments. The failure test is designed to quantify the bending tolerance of motion segments. See the Methods Section at the end of this report for more details on the experimental protocol.

Specimen ID: B35M

Test Reference: B35MO2EXTN

Test Objective: Extension Flexibility

Spine Segment: Occiput - C2

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## Specimen Description

|           |                |         |                 |
|-----------|----------------|---------|-----------------|
| Sex:      | <u>Male</u>    | Age:    | <u>64 years</u> |
| Weight:   | <u>N/A (N)</u> | Height: | <u>N/A (cm)</u> |
| Comments: | No comments.   |         |                 |

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## Video Data:

### Image Data:

Reference Image: F0100.tif

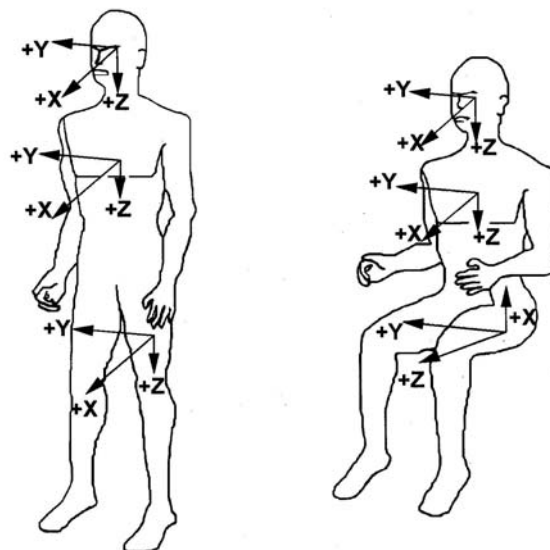
The digital images are of sequential moment load steps for flexion through extension. The angular displacements of the spinal segments are computed relative to the Reference Image. This Reference Image is also used for calculating angular displacements in the failure test.

Test Comments: No comments.

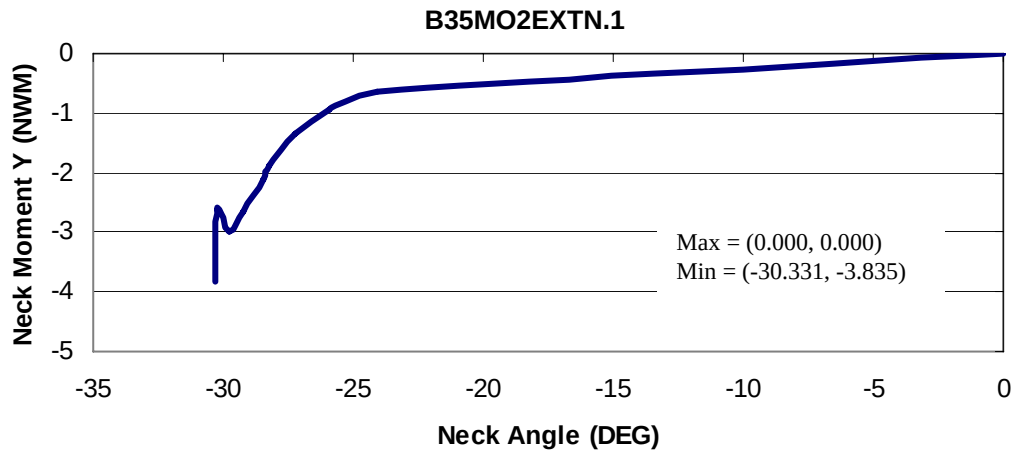
## Appendix A: Coordinate system and sign conventions

Sign Conventions: Compliant with SAE J211/1 March 1995

| Load Cell                         | Measure | Manipulation    | Polarity |
|-----------------------------------|---------|-----------------|----------|
| Denton 2554A Upper Neck Load Cell | My      | Chin to Sternum | +        |



## Appendix C: Computed Data



## Appendix C: Instrumentation

### Instrumentation

| Channel | Transducer                  | Measure      | Manipulation    | Axis |
|---------|-----------------------------|--------------|-----------------|------|
| 1*      | Denton Upper Neck Load Cell | Moment (N-m) | Chin to Sternum | +y   |

### Data Acquisition:

Sampling Frequency: N/A (hz)

Number of Samples: 9

### Computed Data:

| Channel | Computation | Measure   | Manipulation    | Axis |
|---------|-------------|-----------|-----------------|------|
| 2*      | Flexibility | Angle (°) | Chin to Sternum | +y   |

\*Raw data has no time base and is incorporated in the computed flexibility.

## Methods

Unembalmed human heads with intact spines were obtained shortly after death. All specimen handling was performed in compliance with both NHTSA 700-4 guidelines and CDC guidelines. The specimens were sealed in plastic bags, frozen and stored at  $-20^{\circ}$ . All donor medical records were examined to ensure that there were no pre-existing pathologies, such as severe degenerative diseases or spinal pathologies, which could affect the structural responses of the specimens.

The muscular tissues were removed while keeping all the ligamentous structures intact. The lower cervical motion segments (C4-C5, C6-C7) were isolated, cleaned, and cast into aluminum cups with fiber-reinforced polyester resin. The cephalad end of the upper cervical spine specimens was secured using halo fixation. If necessary, the mandible and maxilla were removed in order to allow appropriate flexion range of motion. Upper cervical specimens (Head-C2) were inverted and mounted in the test frame using halo fixation of the head and casting of the C2 vertebra.

The test apparatus applies pure bending moments, and the resulting angular displacements are computed by tracking markers on digital images. The moments are generated using a pair of pneumatic pistons to apply a force-couple. After the specimen was mounted in the test frame, a counterweight was applied to the upper casting assembly. The mass of the counterweight was chosen to counterbalance the mass of the assembly, and to place the motion segment in 0.5 Newton of tension. The purpose of the 0.5 N tensile load was to establish an initial position for the motion segment in the middle of its neutral zone. The specimen was imaged in this position and the load cell values were zeroed. This image served as the reference position for the flexibility and the failure tests. Prior to flexibility testing, the specimens were preconditioned with 30 cycles of  $\pm 1.5$  N-m of moment. The specimens were then loaded with pure flexion and extension moments in 0.5 N-m increments. Thirty seconds of creep were allowed prior to data acquisition, and the load was released between load steps. The peak applied moment was approximately 3.5 N-m. A Denton load cell is placed at the base of the specimen to collect forces and moments and to ensure that the moments remained pure. Lower range six-axis load cell (Denton model 2554A) was used for flexibility tests (FLEX and EXTN) and for failure of lower cervical segments C45 and C67. Higher range three-axis load cell (Denton model 1110AF-5K) was used for upper cervical spine (OC2) failure, which typically occurs at a higher moment than C45 and C67 failures.

After completion of the above bending tests, the tension mass was removed, and the specimens were failed in either flexion or extension. The loading rate was the maximum that could be achieved using our 100 psi system. The rates were dependent on the stiffness of the specimen, and were on the order of 90 N-m/second. The failure tests were imaged at 125 frames/second. Failure was defined as a decrease in the measured moment with increasing rotation, and they were verified by examination of the high-speed images. Specimen dissection was performed to document injuries.

All image data was processed with Matlab. This software enables us to digitize the location of the marker pins in each image. The location of each marker is determined using a centroid algorithm to increase accuracy. Angular displacements for each image are determined with respect to the reference image.